

DAVID PORFIRIO

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RESEARCH INTERESTS

My research intersects **robotics**, **human-computer interaction**, and **artificial intelligence**, in order to design and evaluate innovative technology that enables human-robot task communication. This is important because of the enormous potential that robots have to positively impact daily life. However, this is difficult because robots struggle to capture and understand human needs.

EDUCATION

PhD	University of Wisconsin–Madison (UW–Madison), Madison, WI, USA Computer Sciences	2018-2022
MSc	UW–Madison, Madison, WI, USA Computer Sciences	2016-2018
BS	University of Arizona, Tucson, AZ, USA Double degree (hon) in computer science and physiology Minor in mathematics Summa cum laude	2011-2016

RESEARCH POSITIONS

George Mason University

Assistant Professor (tenure track)
Computer Science Department
College of Engineering and Computing

2025-present

US Naval Research Laboratory

Computer Scientist
Adaptive Systems Section
Navy Center for Applied Research in Artificial Intelligence

2024-2025

NRC Postdoctoral Research Associate

Adaptive Systems Section
Navy Center for Applied Research in Artificial Intelligence
Advisor: Dr. Laura Hiatt

2022-2024

UW–Madison

Doctoral Student
People and Robots Laboratory
Computer Sciences Department
Advisors: Drs. Bilge Mutlu & Aws Albarghouthi
Thesis: *Authoring Social Interactions Between Humans and Robots*
Committee: Drs. Bilge Mutlu, Aws Albarghouthi, Maya Cakmak, & Kevin Ponto

2016-2022

Nokia Bell Labs

Research Intern
New Providence, NJ, USA (Virtual)
Mentors: Drs. Martin Carroll, Kedar Namjoshi, Itai Segall

Summer 2021

University of Arizona

Undergraduate Research Assistant

Department of Computer Science

Advisor: Dr. John Kececioglu

2015-2016

Student, Undergraduate Biology Research Program

Department of Ecology and Evolutionary Biology

Advisor: Dr. Joanna Masel

Summer 2014

Student, Undergraduate Biology Research Program

Department of Physiology

Advisor: Dr. E. Fiona Bailey

2013-2014

CONFERENCE PAPERS

Acceptance rates are reported if known.

1. Zhou, P., Hong, S., & **Porfirio, D.** (2026, November). Attune: A Self-Annotation Tool for Understanding Robot Operator Attention Profiles. In Proceedings of ACM Symposium on User Interface Software and Technology (UIST). In press. Acceptance rate: 20.1%
2. Li, T. & **Porfirio, D.** (2026, June). Distill: Uncovering the True Intent behind Human-Robot Communication. In Proceedings of ACM Designing Interactive Systems (DIS). Acceptance rate: 21%
3. **Porfirio, D.**, McDermott, I., Chen, H., Satake, S., Kanda, T., LaToza, T. (2026, June). Designing for Robot Wranglers: A Conceptual Framework Grounded in Literature and Practice. In Proceedings of ACM Designing Interactive Systems (DIS). Acceptance rate: 21%
4. **Porfirio, D.**, Hsiao, V., Fine-Morris, M., Smith, L., & Hiatt, L. (2025, August). Bootstrapping Human-Like Planning via LLMs. In Proceedings of the 2025 International Conference on Robot and Human Interactive Communication (RO-MAN). Acceptance rate: 60%
5. **Porfirio, D.**, Roberts, M., & Hiatt, L. (2025, May). Uncertainty Expression for Human-Robot Task Communication. In Proceedings of the 24th International Conference on Autonomous Agents and Multi-agent Systems (AAMAS). Acceptance rate: 24.5%
6. Lee, C., **Porfirio, D.**, Wang, J., Zhao, K., & Mutlu, B. (2025, April). VeriPlan: Integrating Formal Verification and LLMs into End-User Planning. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI). Acceptance rate: 25.1%
7. **Porfirio, D.**, Roberts, M., & Hiatt, L. M. (2025, March). An Interaction Specification Language for Robot Application Development. In 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI). **Best Short Contribution Award** Acceptance rate: 43.5%
8. Stegner, L., Hwang, Y., **Porfirio, D.**, & Mutlu, B. (2024, July). Understanding On-the-Fly End-User Robot Programming. In Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS). Acceptance rate: 25%
9. **Porfirio, D.**, Roberts, M., & Hiatt, L. (2024, March). Goal-Oriented End-User Programming of Robots. In Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI). **NRC/ASEE Postdoctoral Research Publications Award** Acceptance rate: 25%
10. **Porfirio, D.**, Stegner, L., Cakmak, M., Sauppé, A., Albarghouthi, A., & Mutlu, B. (2023, March). Sketching Robot Programs On the Fly. In 2023 18th ACM/IEEE International Conference on Human-Robot Interaction (HRI). ACM/IEEE. Acceptance rate: 25%
11. **Porfirio, D.**, Stegner, L., Cakmak, M., Sauppé, A., Albarghouthi, A., & Mutlu, B. (2021, May). Figaro: A Tabletop Authoring Environment for Human-Robot Interaction. In Proceedings of the 2021 Conference on Human Factors in Computing Systems (CHI). Acceptance rate: 26%

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| 12. | Porfirio, D. , Sauppé, A., Albarghouthi, A., & Mutlu, B. (2020, April). Transforming robot programs based on social context. In Proceedings of the 2020 conference on human factors in computing systems (CHI). | Acceptance rate: 24% |
| 13. | Porfirio, D. , Fisher, E., Sauppé, A., Albarghouthi, A., & Mutlu, B. (2019, October). Bodystorming human-robot interactions. In Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST). | Acceptance rate: 24% |
| 14. | Porfirio, D. , Sauppé, A., Albarghouthi, A., & Mutlu, B. (2018, October). Authoring and verifying human-robot interactions. In The 31st Annual ACM Symposium on User Interface Software and Technology (UIST) Best Paper Award | Acceptance rate: 21% |

JOURNAL PAPERS

1. Hwang, Y., Lee, C., **Porfirio, D.**, Hiatt, L. M., and Mutlu, B., “Formal Methods in Robot End-User Development: Progress, Gaps, and Opportunities”. Foundations and Trends in Robotics. In press.
2. Xiong, K., McEntee, J. P., **Porfirio, D. J.**, & Masel, J. (2017). Drift barriers to quality control when genes are expressed at different levels. Genetics, 205(1), 397-407.
3. Shumway, K. R., **Porfirio, D. J.**, & Bailey, E. F. (2015). Phonation-related rate coding and recruitment in the genioglossus muscle. Experimental brain research, 233(7), 2133-2140.

WORKSHOP PAPERS, COMPANION PROCEEDINGS, AND SPECIAL SESSION PAPERS

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| 1. | Tiger, J., Williamson, P., Kannan, N., Henley, J., Roesler, E., & Porfirio, D. (2026, March). Nose Knows: A Human-Centered Robotic Air Quality Monitor. In 2026 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI). Honorable Mention | Acceptance rate: 49.1% |
| 2. | Sullivan, D., Porfirio, D. , Mutlu, B., & Hiatt, L. (2026, March). Examining the Effectiveness of Obfuscatory Planning Strategies Through Human Observation. In 2026 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI). Best Late-Breaking Report Nominee | |
| 3. | Praveena, P., Schoen, A., Gleicher, M., Porfirio, D. , & Mutlu, B. (2023, October). Petri Nets for the Iterative Development of Interactive Robotic Systems. In 2023 AAAI Fall Symposium Series on <i>Unifying Representations for Robot Application Development</i> (UR-RAD). | |
| 4. | Stegner, L., Porfirio, D. , Roberts, M., & Hiatt, L. (2023, October). Considerations for End-User Development in the Caregiving Domain. In 2023 AAAI Fall Symposium Series on <i>Unifying Representations for Robot Application Development</i> (UR-RAD). | |
| 5. | Porfirio, D. , Roberts, M., & Hiatt, L. (2023, August). Guidelines for a Human-Robot Interaction Specification Language. In 2023 IEEE International Symposium on Robot and Human Interactive Communication Special Session on <i>HRI in Academia and Industry: Bridging the Gap</i> . IEEE. | |
| 6. | Porfirio, D. , Roberts, M., & Hiatt, L. (2023, March). On a Standardized Logical Representation for Human-Robot Interaction. In 2023 AAAI Spring Symposium, <i>HRI in Academia and Industry: Bridging the Gap</i> . Best Paper Nominee | |
| 7. | Porfirio, D. , Sauppé, A., Cakmak, M., Albarghouthi, A., & Mutlu, B. (2023, March). Crowdsourcing Task Traces for Service Robotics. In 2023 18th ACM/IEEE International Conference on Human-Robot Interaction (HRI). | |
| 8. | Pelikan, H., Porfirio, D. , Winkle, K. (2023, March). Designing Better Human-Robot Interactions through Enactment, Engagement, and Reflection. In 2023 18th ACM/IEEE International Conference on Human-Robot Interaction Workshop on <i>Human-Robot Conversational Interaction</i> (CUI@HRI). | |

9. **Porfirio, D.**, Cakmak, M., Saup  , A., Albarghouthi, A., & Mutlu, B. (2021, May). Interaction Templates: A Data-Driven Approach for Authoring Robot Programs. In 2021 12th Annual Workshop on Evaluation and Usability of Programming Languages and Tools (PLATEAU).
10. **Porfirio, D.**, Saup  , A., Albarghouthi, A., & Mutlu, B. (2019, March). Computational Tools for Human-Robot Interaction Design. In 2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI) (pp. 733-735). IEEE. Acceptance rate: 31%

GRANTS AND GIFTS

Jerome and Isabella Karle Distinguished Scholar \$25,000—seed funding from the U.S. Naval Research Laboratory	2024-2025
Microsoft Dissertation Grant \$21,148—unrestricted gift from Microsoft	2021
Quori Robot Program Robot donation—received <i>Quori</i>	2018
Galileo Circle Scholar \$1,000—unrestricted gift from UA donors	2015

FELLOWSHIPS

Jerome and Isabella Karle Distinguished Scholar Supports a full-time appointment at the U.S. Naval Research Laboratory	2024-2025
NRC Postdoctoral Research Fellowship Stipend with travel support at the U.S. Naval Research Laboratory	2022-2024
Cisco Graduate Student Fellowship Selected by the UW–Madison Computer Sciences Department	2021-2022
NSF Graduate Research Fellowship Stipend at the UW–Madison Computer Sciences Department	2017-2022
Advanced Opportunity Fellowship Selected by the UW–Madison Computer Sciences Department	2016, 2020

HONORS AND AWARDS

Honorable Mention for Student Design Competition — <i>Faculty Mentor</i> HRI '26: <i>Nose Knows: A Human-Centered Robotic Air Quality Monitor.</i>	2026
Best Late-Breaking Report Nominee — <i>Co-author</i> HRI '26: <i>Examining the Effectiveness of Obfuscatory Planning Strategies Through Human Observation.</i>	2026
NRC/ASEE Postdoctoral Research Publications Award — <i>Lead Author</i> HRI '24: <i>Goal-Oriented End-User Programming of Robots</i>	2025
Best Paper Award — <i>Lead Author</i> HRI '25: <i>An Interaction Specification Language for Robot Application Development</i>	2025
Best Paper Nominee — <i>Lead Author</i> AAAI Spring Symposium '23: <i>On a Standardized Logical Representation for Human-Robot Interaction</i>	2023
Best Paper Award — <i>Lead Author</i> UIST '18: <i>Authoring and Verifying Human-Robot Interactions</i>	2018

Excellence in Undergraduate Research Award Selected by the UA Department of Computer Science	2016
Galileo Circle Scholar Selected by the UA Department of Computer Science	2015
National Hispanic Scholar Selected by the National Hispanic Recognition Program	2011
Dean's List with Distinction Awarded during six semesters at UA	2011-2016

TEACHING EXPERIENCE

George Mason University <i>CS 685, Autonomous Robotics</i> Description: Theories and practices of autonomous robotics.	Fall 2026
<i>CS 690/ME 585/ECE 585, Human-Robot Interaction</i> Description: Principles and methods of human-robot interaction.	Spring 2026
<i>CS 690-007, Evaluating Generative AI Systems (course I designed)</i> Description: An empirical look at systems that use generative artificial intelligence.	Fall 2025
Guest Lecturer <i>CMSC 722, AI Planning. University of Maryland, College Park.</i>	December 2025
<i>CSCI 436/536, Human-Robot Interaction. Colorado School of Mines.</i>	November 2025
<i>ME 585, Human-Robot Interaction. George Mason University.</i>	October 2025
<i>CMSC 722, AI Planning. University of Maryland, College Park.</i>	December 2024
Grandparents University Instructor, UW <i>Social Robotics (lab sessions I co-designed)</i> Description: two-day lecture and lab sessions for UW alumni families.	Summers 2018-2019
Teaching Assistant, UA <i>CSC 352, Systems Programming and Unix. Summer 2015.</i>	Summer 2015
Section Leader, UA <i>CSC 127B, Introduction to Computer Science II.</i>	Spring 2015
<i>CSC 245, Introduction to Discrete Structures.</i>	Fall 2015
<i>CSC 227, Program Design and Development.</i>	Fall 2014.
Student Preceptor, UA <i>CHEM 151, Chemical Thinking I. Spring 2012.</i>	Spring 2012

INVITATIONS

George Mason University AI Club—Speaker Title: <i>Bidirectional Human-Robot Task Communication</i>	April 2026
HRI Pioneers 2026—Panelist	March 2026
George Mason University—Speaker Title: <i>End-User Development for Human Robot Interaction: A Shifting Paradigm</i>	April 2025
HRI Workshop on The Road to Reliable Robots—Speaker Title: <i>Evaluating Systems for Robot Application Development</i>	March 2025
Johns Hopkins University—Speaker Title: <i>Robot Application Development: A Shifting Paradigm</i>	April 2024

Colorado School of Mines — <i>Speaker</i> Title: <i>Authoring Social Interactions Between Humans and Robots</i>	March 2022
University at Buffalo — <i>Speaker</i> Title: <i>Authoring Social Interactions Between Humans and Robots</i>	February 2022
Montana State University — <i>Speaker</i> Title: <i>Authoring Social Interactions Between Humans and Robots</i>	February 2022
Talking Robotics — <i>Speaker</i> Title: <i>Authoring Social Interactions Between Humans and Robots</i>	December 2020
UW–Madison, CS Student Research Symposium — <i>Speaker</i> Title: <i>Applying Formal Methods to Human-Robot Interaction</i>	April 2019
Other events invited to attend	
2022—Robotics Perception and Learning Summer School, KTH Royal Institute of Technology	
2020—Microsoft Research AI Breakthroughs	
2019—Heidelberg Laureate Forum	
2019—HRI Pioneers 2018—UIST School	

ADVISING AND MENTORSHIP

Ting Li — <i>Advisee</i> Ph.D. Student, George Mason University	2025-present
Puqi Zhou — <i>Advisee</i> Ph.D. Student, George Mason University	2025-present
Chase Morgan — <i>Advisee</i> PhD Student, George Mason University	2026-present
Celine Mang — <i>Advisee</i> PhD Student, George Mason University	2026-present
Zhihao Zhang — <i>Advisee</i> PhD Student, George Mason University	2026-present
Ryan Ferzoco — <i>Advisee</i> MS Student, George Mason University	2026-present
Alex Veerasammy — <i>Advisee</i> MS Student, George Mason University	2026-present
Other Students Advised or Mentored	
Yazan Alshoroogi (2025)— <i>George Mason University</i>	
Evan Conway (2024-2025)— <i>US Naval Research Laboratory</i>	
Nhi Tran (2023-2025)— <i>US Naval Research Laboratory</i>	
Madeline Forsythe (2024)— <i>US Naval Research Laboratory</i>	
Kyle Wang (2019-2022)— <i>UW–Madison</i>	
Ali Zaidi (2017-2020)— <i>UW–Madison</i>	
Linda Wu (2019-2020)— <i>UW–Madison</i>	
Mikayla Buford (2019-2020)— <i>UW–Madison</i>	
Laik Ruetten (2020)— <i>UW–Madison</i>	
Akshat Khanna (2018-2019)— <i>UW–Madison</i>	
Zhechun Zhou (2019)— <i>UW–Madison</i>	
Arabella Yao (2019)— <i>UW–Madison</i>	
Ezra Boley (2019)— <i>UW–Madison</i>	
Evan Fisher (2018)— <i>UW–Madison</i>	
Pranav Rajiv (2017-2018)— <i>UW–Madison</i>	

Yue Sun (2017)—*UW-Madison*
Raghav Bagwat (2017)—*UW-Madison*
Sherine Zhang (2017)—*UW-Madison*
Jack Weissburg (2017)—*UW-Madison*

Doctoral Committee Member for the Following Students

Saad Elbeleidy (2024)—*Colorado School of Mines*

EXTERNAL SERVICE

HRI 2027

Co-organizer—*registration co-chair*
Program committee member—*technical track*

UR-RAD 2026

Co-organizer

DIS 2026

Session Chair—*AI in Education and Reflection*

HRI 2026

Session Chair—*Robot Perception 2*
Program committee member—*systems track*

UR-RAD 2025—*AAAI Fall Symposium*

Co-chair

HRI 2025

Program committee member—*systems track*

UR-RAD 2024—*AAAI Fall Symposium*

Chair

NordiCHI 2024 Tutorial—*Designing Human-Robot Interactions: A StEER Tutorial*

Co-organizer

HRI 2024

Program committee member—*systems track*

HRI 2024 Workshop—*End-User Development for Human-Robot Interaction*

Co-chair

UR-RAD 2023—*AAAI Fall Symposium*

Chair

HRI 2021 Workshop—*Participatory Design and End-User Programming for Human-Robot Interaction*

Co-organizer

Paper Reviewing

HRI	-	ACM/IEEE International Conference on Human-Robot Interaction	2020-2026
CHI	-	ACM Conference on Human Factors in Computing Systems	2022, 2024-2026
AAAI	-	Annual AAAI Conference on Artificial Intelligence	2026
AAMAS	-	International Conference on Autonomous Agents and Multiagent Systems	2024-2026
CogSci	-	Annual Meeting of the Cognitive Science Society	2024-2026
RSS	-	Robotics: Science and Systems	2025-2026
ICRA	-	IEEE International Conference on Robotics and Automation	2024-2025
THRI	-	ACM Transactions on Human-Robot Interaction	2020, 2024-2025
HAI	-	International Conference on Human-Agent Interaction	2023, 2025
UIST	-	ACM Symposium on User Interface Software and Technology	2021, 2025
RAL	-	Robotics and Automation Letters	2025
ECAI	-	European Conference on Artificial Intelligence	2025
SORO	-	International Journal of Social Robotics	2022, 2024
CSCW	-	ACM SIGCHI Conference on Computer-Supported Cooperative Work	2022
TAHRI	-	International Symposium on Technological Advances in Human-Robot Interaction	2024
AI-HRI	-	AAAI Fall Symposium on Artificial Intelligence for Human-Robot Interaction	2021

Grant Reviewing

Army Research Office - 2026

Icelandic Research Fund - 2026